Tech Report Topic Guidelines **Physics 50**

*The purpose of the tech report is to give you experience in defining a question, carrying out a procedure to address that question efficiently and quantitatively, and explaining the entire logical sequence in an oral presentation.

**The tech report is not a senior thesis.* You have six hours in lab, and you may have to share the apparatus with another group. Choose a specific, relatively simple question that can be posed in quantitative terms and answered via an experiment you will have time to carry out. Your question can be based on a problem you encountered when doing the experiment for the first time (perhaps your improvements could turn into the standard lab procedure in future years!). It can be a detailed look at one small aspect of the original experiment or a totally new question investigated on the original apparatus, but make sure you will not run out of time if you take this approach.

<u>Some ideas if you need inspiration:</u>

Module 1: Acoustic Levitator and Speed of Sound

*Can you do the measurement better using other tools (hardware or software)?

*Measure the speed of sound in a totally different way. (We have microphones and oscilloscopes so you can measure the time interval between a sharp noise reaching two different microphones. You can also try measuring the speed of sound outdoors by creating a sound and a light at the same time and measuring the delay between arrival of those two signals at a known distance or distances away.)

*Measure speed of waves on a string at various tensions. (We have an apparatus that excites vibrations at a series of low-ish frequencies so you can view the standing wave and measure its wavelength directly.)

Module 2: Diffraction

*Can you do the Week 2 measurement(s) better using other tools or procedures?

*How does the reflected diffraction pattern from the LCD panel compare to the transmitted diffraction pattern?

*Use single-slit diffraction to measure the width of human hairs.

*Measure pixel structure/spacing of your phone.

*Measure spacing between grooves on a CD or DVD.

*Measure diffraction through a fabric like tulle as you stretch it.

*Measure diffraction through a colloidal crystal oriented at different angles.

*Imitate Rosalind Franklin's X-ray diffraction experiments on DNA, but using a small spring as your helix and a red laser instead of X-rays.

Solar Panels

*Investigate an aspect of solar panel performance based on the preliminary results found in the Experimental Design lab.

*What happens if you try to hook up two solar panels in series or in parallel?

*How does solar panel performance change with time of day or with season (real or simulated)?